

# Product Requirements Document (PRD)

## Product Name

**StadiumAI – GenAI-Powered Stadium Operations & Fan Experience Platform**

**Event:** FIFA World Cup 2026

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## 1. Product Vision

StadiumAI is an AI-powered platform that enhances stadium operations and delivers a seamless, personalized experience for every fan attending FIFA World Cup 2026 matches.

The platform combines Generative AI, real-time analytics, IoT sensors, and operational intelligence to assist stadium personnel, improve safety, reduce operational delays, and provide personalized digital assistance throughout the event.

The solution enables proactive decision-making by transforming live stadium data into actionable recommendations for staff while delivering instant guidance and engagement for attendees.

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## 2. Problem Statement

Large sporting events face several operational challenges, including:

- Long entry and concession queues
- Difficulty managing crowd movement
- Delayed incident response
- Communication gaps between teams
- Limited personalized assistance for visitors
- High operational workload during peak periods

These issues reduce operational efficiency and negatively impact the fan experience.

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## 3. Product Goals

### Operational Goals

- Improve stadium efficiency
- Reduce emergency response time

- Optimize workforce deployment
- Minimize crowd congestion
- Improve operational visibility

### **Fan Experience Goals**

- Reduce waiting times
  - Provide personalized navigation
  - Deliver real-time match updates
  - Offer multilingual AI assistance
  - Increase fan engagement
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## **4. User Personas**

### **4.1 Stadium Staff**

#### **Responsibilities**

- Crowd monitoring
- Security
- Maintenance
- Entry management
- Incident handling

#### **Needs**

- Real-time alerts
  - Location-based incident notifications
  - AI-generated recommendations
  - Fast communication
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### **4.2 Event Organizers**

#### **Responsibilities**

- Stadium operations
- Resource management
- Performance monitoring
- Decision making

#### **Needs**

- Live dashboards
- Crowd analytics

- Staff utilization reports
  - Operational KPIs
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## 4.3 Fans

### Needs

- Easy stadium navigation
  - Seat guidance
  - Queue information
  - Food recommendations
  - Match updates
  - Emergency assistance
  - Personalized notifications
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## 5. User Journey Examples

### Stadium Staff

1. Receive AI alert for Gate B congestion.
  2. View recommended response.
  3. Redirect nearby staff.
  4. Monitor improvement.
  5. Close incident.
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### Event Organizer

1. Open Operations Dashboard.
  2. Review crowd density.
  3. View security alerts.
  4. Deploy resources.
  5. Monitor KPIs.
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### Fan

1. Open StadiumAI App.
2. Scan ticket.
3. Receive navigation to seat.
4. Get shortest concession queue.
5. Receive live match notifications.

6. Receive exit recommendations after match.

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## **6. Core Features**

### **6.1 AI Stadium Assistant**

- Natural language chatbot
  - Voice interaction
  - Multilingual support
  - FAQs
  - Navigation assistance
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### **6.2 Real-Time Alerts**

Examples:

- Medical emergency
  - Crowd congestion
  - Unauthorized access
  - Lost child
  - Equipment malfunction
  - Weather alerts
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### **6.3 Crowd Management**

- Heatmap visualization
  - Predictive congestion detection
  - Route optimization
  - Entry balancing
  - Exit recommendations
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### **6.4 Personalized Fan Experience**

- Personalized notifications
  - Seat navigation
  - Parking guidance
  - Merchandise recommendations
  - Food suggestions
  - Nearby facility recommendations
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## 6.5 Operations Dashboard

Displays:

- Live attendance
  - Queue length
  - Security incidents
  - Resource allocation
  - Crowd heatmaps
  - Response times
  - AI insights
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## 6.6 Emergency Response

- AI incident detection
  - Automatic staff assignment
  - Live location sharing
  - Emergency broadcast
  - Evacuation guidance
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## 6.7 Fan Engagement

- Live polls
  - Trivia
  - Match statistics
  - AI-generated highlights
  - Loyalty rewards
  - Digital collectibles
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# 7. Functional Requirements

### FR-01

Provide AI chatbot assistance.

### FR-02

Generate live operational alerts.

### FR-03

Track crowd density.

**FR-04**

Support multilingual communication.

**FR-05**

Recommend optimal navigation.

**FR-06**

Provide personalized recommendations.

**FR-07**

Support emergency notifications.

**FR-08**

Display operational analytics.

**FR-09**

Integrate with ticketing systems.

**FR-10**

Provide administrator controls.

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## 8. Non-Functional Requirements

- High availability
  - Low latency
  - Secure communication
  - Mobile responsive
  - Accessibility compliant
  - Easy maintenance
  - Fault tolerant
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## 9. Success Metrics

### Operational

- 25% reduction in queue time

- 40% faster incident response
  - 30% reduction in crowd congestion
  - 99.95% platform availability
  - Less than 2-second dashboard refresh
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## Fan Experience

- 90% chatbot satisfaction
  - 95% navigation accuracy
  - 85% user engagement
  - App rating above 4.5/5
  - 30% increase in digital engagement
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## 10. Risks

- Network outages
  - Peak traffic spikes
  - Sensor failures
  - AI hallucinations
  - Privacy concerns
  - Hardware failures
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## 11. Future Enhancements

- AI-powered autonomous surveillance
  - Digital twin stadium simulation
  - AR indoor navigation
  - Predictive maintenance
  - Emotion analysis
  - Smart parking optimization
  - Wearable device integration
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## 12. Product Summary

StadiumAI combines Generative AI, IoT, and real-time analytics into a unified platform that improves operational efficiency, enhances fan engagement, and strengthens safety. The solution supports stadium staff with intelligent decision-making while providing attendees with personalized, multilingual assistance throughout FIFA World Cup 2026 events.